Circular Economy Improvement Using Machine Learning

Classifying Products as Recyclable or Biodegradable Through Image Analysis

(PROJECT PHASE- I)
submitted in partial fulfillment of the requirements
for the award of the degree in

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING

by

HARICHSELVAM (Reg.No:211191101045)



DEPARTMENT
OF
COMPUTER SCIENCE AND ENGINEERING

SEPTEMBER 2024

DECLARATION FORMAT

I C.HARICHSELVAM(Reg.No:211191101045), hereby declare that the Project Report (ProjectPhase-I)entitled"**Circular Economy Improvement Using Machine Learning**" is done by us under the guidance of Dr./Prof. T.V.Ananthan is submitted in partial fulfillment of the requirements for the award of the degree in BACHELOR OF TECHNOLOGY in Computer Science and Engineering.

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BONAFIDE CERTIFICATE

This is to certify that this Project Report (Project Phase-I) is the bonafide work of Mr. Harichselvam.C Reg. No:21119110145, who carried out the project entitled "**Circular Economy Improvement Using Machine Learning**" under our supervision from June 2024 to Sep 2024.

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CONTENTS

Section	Page Number
Abstract	7
Chapter 1: Introduction	8
1.1 Understanding Economy Types	
1.1.1 Traditional Economy	
1.1.2 Circular Economy	
1.1.3 The Need for a Shift	
Chapter 2: Requirement Analysis	10
2.1 Problem Domain	
2.1.1 Increasing Waste Mismanagement	
2.1.2 Inadequate Recycling Processes	
2.1.3 Technological Limitations	
2.1.4 Need for Public Participation	
Chapter 3: Requirement Specification	11
3.1 User Requirements	
3.2 System Requirements	
3.3 Development and Runtime Requirements	
Chapter 4: Design	12
4.1 Architectural Design	
4.1.1 System Architecture	
4.1.2 Module Interaction	
4.2 Database Design	
4.3 User Interface Design	
Chapter 5: Implementation	17
5.1 Backend Development	
5.1.1 Flask Framework	
5.1.2 RESTful APIs	
5.2 Machine Learning Model	
5.2.1 Dataset Preparation	
5.2.2 Model Architecture	
5.2.3 Training and Validation	
5.3 Database Management	
Chapter 6: Testing and Evaluation	20
6.1 Testing Strategies	
6.2 Performance Evaluation	
6.3 User Feedback	
Chapter 7: Conclusion and Future Work	22
Bibliography/References/Appendix	

LIST OF ABBREVIATIONS

- **AI**: Artificial Intelligence

- **CNN**: Convolutional Neural Network
- **CSS**: Cascading Style Sheets
- **HTML**: Hypertext Markup Language
- **JSON**: JavaScript Object Notation
- **ML**: Machine Learning
- **NoSQL**: Non-Structured Query Language
- **REST**: Representational State Transfer
- **UI**: User Interface

ABSTRACT

OBJECTIVE:

To innovatively design, develop, and deploy a comprehensive intelligent system that harnesses advanced machine learning techniques—specifically Convolutional Neural Networks (CNNs)—to accurately and efficiently classify images of waste products uploaded by users.

This system will be accessible through a seamless and user-friendly web application developed using Flask, allowing users to interact with the platform effortlessly. The application will process user-uploaded images in real-time, swiftly identifying and classifying waste items as either recyclable or biodegradable with high precision.
The backend infrastructure will integrate MongoDB for robust and efficient data storage and management, ensuring scalability and reliability. The machine learning models will be developed and trained using TensorFlow and Keras, leveraging their powerful computational capabilities and extensive libraries for optimizing model performance. By automating the waste classification process, the system aims to significantly reduce human error associated with manual sorting, thereby increasing the efficiency and effectiveness of waste management practices.

Furthermore, the project seeks to elevate recycling rates by providing immediate feedback to users, educating them on proper waste disposal methods, and promoting environmentally responsible behaviors. It supports environmental sustainability efforts by contributing to the circular economy model—extending the lifecycle of products and ensuring that materials are appropriately recycled or decomposed. Through this intelligent system, we aim to foster resource efficiency, environmental stewardship, and public awareness, ultimately contributing to a more sustainable and eco-friendly society.

CHAPTER 1: INTRODUCTION

1.1 UNDERSTANDING ECONOMY TYPES

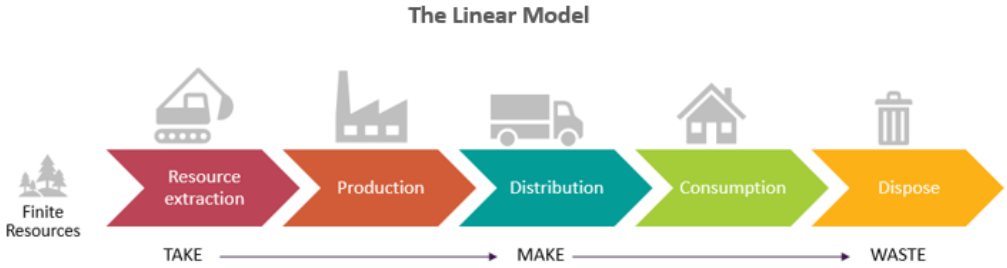
1.1.1 TRADITIONAL ECONOMY

The traditional economy is characterized by a linear model of production and consumption, commonly referred to as **'take-make-dispose'**. In this model, natural resources are extracted (take), transformed into products (make), and eventually discarded as waste after use (dispose).

This approach leads to:

- **Resource Depletion:** Continuous extraction of finite resources without sustainable practices leads to scarcity.
- **Environmental Degradation:** Disposal of waste contributes to pollution, affecting land, air, and water quality.
- **Economic Inefficiency:** Valuable materials are lost in landfills instead of being recovered and reused.

Figure 1.1: Linear Economy Model



(Fig 1.1: A diagram illustrating the linear economy model.)

1.1.2 CIRCULAR ECONOMY

In contrast, the **circular economy** is an innovative model designed to redefine growth by focusing on positive society-wide benefits.

It involves:

- **Resource Efficiency:** Maximizing the utility of resources by keeping them in use for as long as possible.
- **Waste Reduction:** Designing out waste and pollution through sustainable product design and manufacturing processes.
- **Regeneration of Natural Systems:** Returning valuable nutrients to the environment and enhancing ecosystem resilience.

Key principles include:

- **Design for Durability:** Products are designed to last longer, reducing the need for frequent replacements.
- **Reuse and Repair:** Encouraging repairing products instead of disposing of them.
- **Recycling and Upcycling:** Converting waste materials into new, useful products.

Figure 1.2: Circular Economy Model



(Fig 1.2: A diagram illustrating the circular economy model.)

1.1.3 THE NEED FOR A SHIFT

The shift from a traditional to a circular economy is imperative due to:

- **Environmental Concerns:** Addressing climate change, pollution, and loss of biodiversity.
- **Economic Opportunities:** Creating new business models, jobs, and economic growth through sustainable practices.
- **Social Benefits:** Improving quality of life by promoting healthier environments and sustainable consumption patterns.

CHAPTER 2: REQUIREMENT ANALYSIS

2.1 PROBLEM DOMAIN

2.1.1 INCREASING WASTE MISMANAGEMENT

Globally, waste generation is projected to increase by 70% by 2050 unless urgent action is taken (World Bank, 2018).

Mismanagement includes:

- **Improper Segregation:** Mixing of recyclable and non-recyclable waste.
- **Limited Infrastructure:** Insufficient facilities for waste processing and recycling.
- **Lack of Awareness:** Public unawareness about proper waste disposal practices.

2.1.2 INADEQUATE RECYCLING PROCESSES

Challenges in current recycling systems:

- **Inefficient Sorting:** Manual sorting is time-consuming and prone to errors.
- **Contamination:** Non-recyclable materials contaminate recyclable waste streams.
- **Economic Constraints:** Recycling processes can be cost-prohibitive without economies of scale.

2.1.3 TECHNOLOGICAL LIMITATIONS IN WASTE SORTING

Limitations include:

- **Lack of Automation:** Reliance on manual labor reduces efficiency.
- **Insufficient Technology Adoption:** Advanced technologies like AI and robotics are underutilized.
- **Scalability Issues:** Existing solutions may not handle large volumes effectively.

2.1.4 NEED FOR ENHANCED PUBLIC PARTICIPATION IN RECYCLING

To improve recycling rates:

- **Engagement Tools:** Providing accessible platforms for the public to participate.
- **Education and Awareness:** Informing communities about the benefits of recycling.
- **Incentive Programs:** Encouraging participation through rewards or recognition.

CHAPTER 3: REQUIREMENT SPECIFICATION

3.1 USER REQUIREMENTS

- **Easy Image Upload:** Users can upload images of waste products using a user-friendly interface.
- **Immediate Feedback:** The system provides real-time classification results.
- **Educational Resources:** Access to information on proper disposal methods for different waste types.
- **Privacy Assurance:** User data and images are securely handled.

3.2 SYSTEM REQUIREMENTS

- **High Accuracy:** The ML model should achieve high accuracy in classifying waste types.
- **Scalability:** The system must handle multiple concurrent users without performance degradation.
- **Robustness:** Ability to handle various image qualities and backgrounds.
- **Integration Capability:** Seamless integration with web applications and databases.
- **Hardware Requirements**:
  - **Processor**: Intel i5 or higher
  - **RAM**: 16 GB or more
  - **Storage**: At least 500 GB HDD or SSD
  - **Graphics**: Optional but recommended for model training (e.g., NVIDIA GPU)

3.3 DEVELOPMENT AND RUNTIME REQUIREMENTS

- **Programming Language: Python**
  - Used for backend development and machine learning due to its simplicity and extensive libraries.
- **Frameworks:**
  - **Flask:** A lightweight web framework for building responsive web applications, handling HTTP requests, and managing sessions.
  - **TensorFlow and Keras:** Libraries for developing and deploying Convolutional Neural Networks (CNNs) for image classification, with support for efficient training on large datasets.
- **Database: MongoDB**
  - A NoSQL database ideal for storing diverse waste classification data and user-uploaded images, allowing flexible data storage.
- **Frontend Technologies:**
  - HTML5, CSS3, and Bootstrap: HTML5 for structure, CSS3 for styling, and Bootstrap for responsive design, ensuring a mobile-friendly interface.
- **Environment:**
  - Compatible with major operating systems (Windows, Linux, macOS), enabling flexibility in development and deployment.

CHAPTER 4: DESIGN

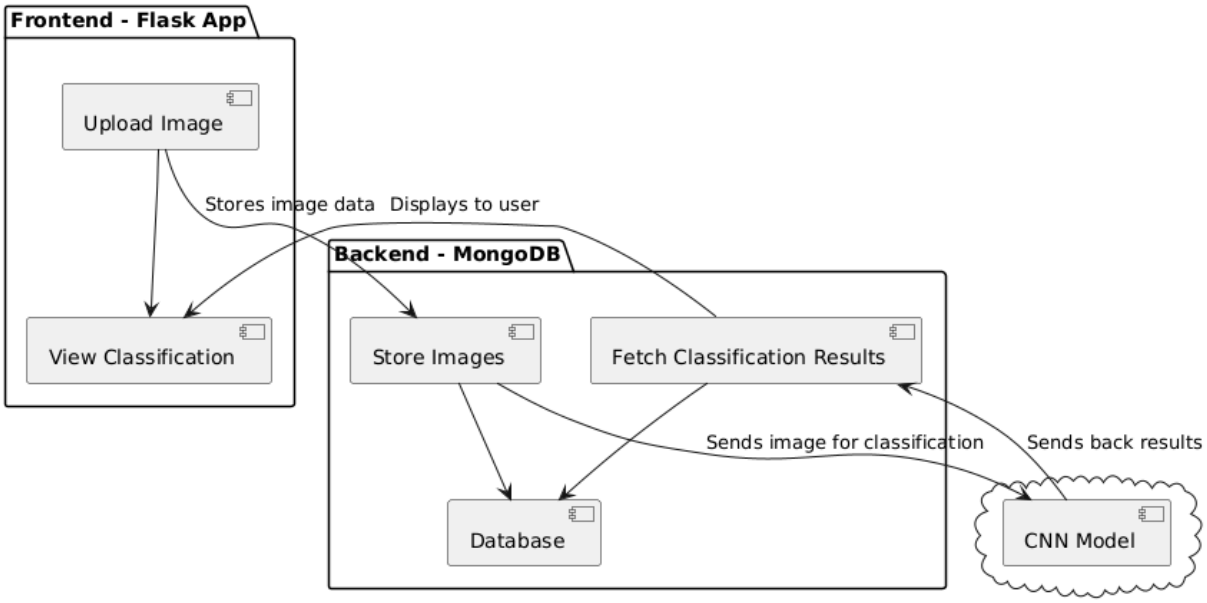
4.1 ARCHITECTURAL DESIGN

4.1.1 SYSTEM ARCHITECTURE

The system follows a **client-server architecture** with the following components:

- **Client Side:**
  - User interface for image upload and displaying results.
  - Responsive design compatible with various devices.
- **Server Side:**
  - **Flask Application:** Handles HTTP requests and routes.
  - **Machine Learning Module:** Processes images and predicts classifications.
  - **Database Layer:** Stores user data, images, and results.

FIGURE 4.1: SYSTEM ARCHITECTURE DIAGRAM



(Fig 4.1: Diagram showing the interaction between client, server, ML model, and database.)

4.1.2 MODULE INTERACTION

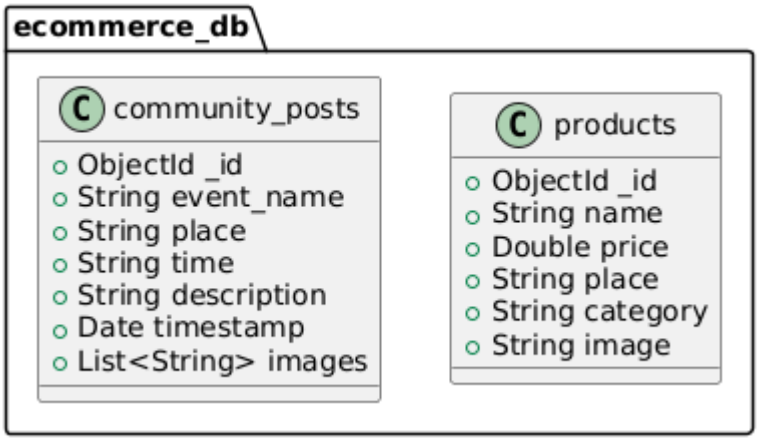
- **User Interface Module:** Captures user input and displays outputs.
- **Controller Module:** Manages application logic and workflows.
- **ML Module:** Performs image processing and classification.
- **Database Module:** Handles data persistence and retrieval.

4.2 DATABASE DESIGN

- **Collections:**
  - **Users:** Stores user credentials and profiles (if user accounts are implemented).

- **Images:** Stores base64-encoded images and associated metadata.
- **Classifications:** Records of classification results and timestamps.

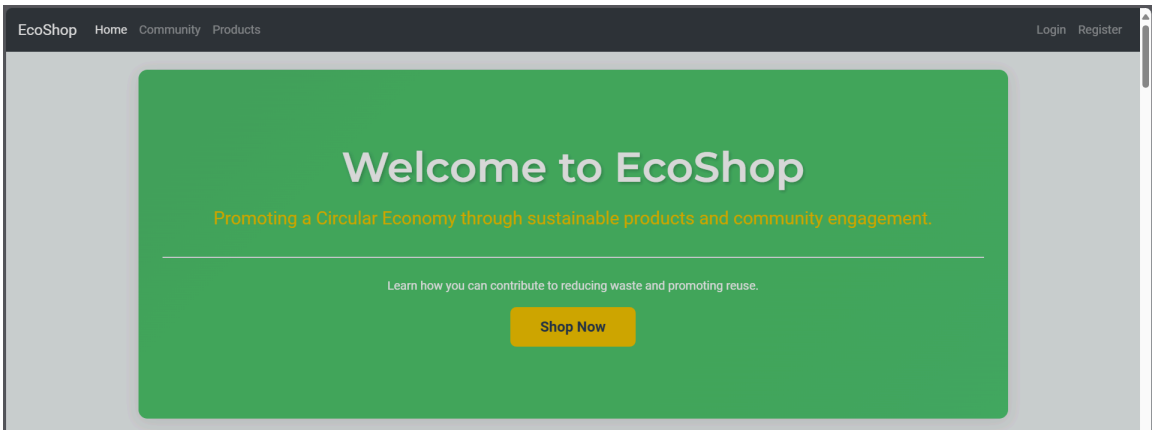
Figure 4.2: Database Schema Diagram

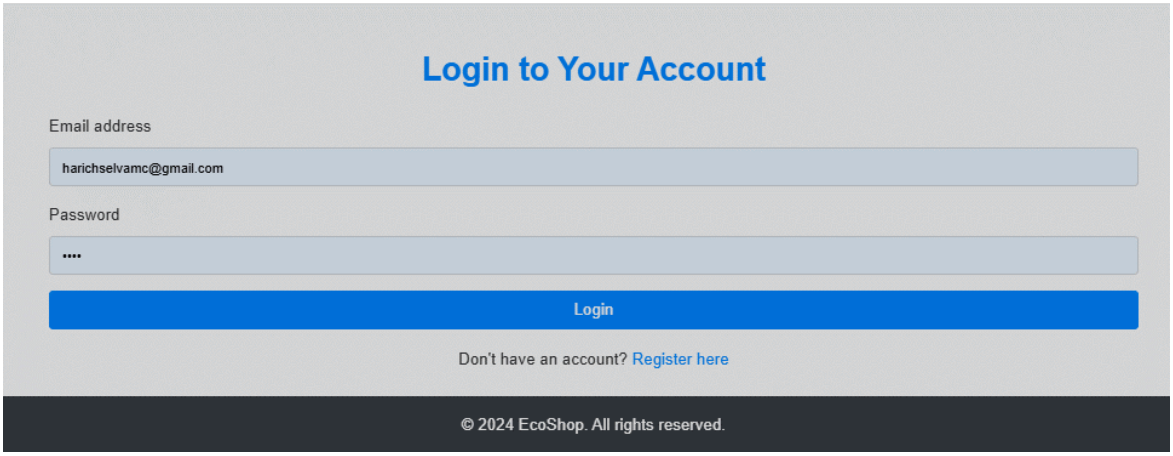


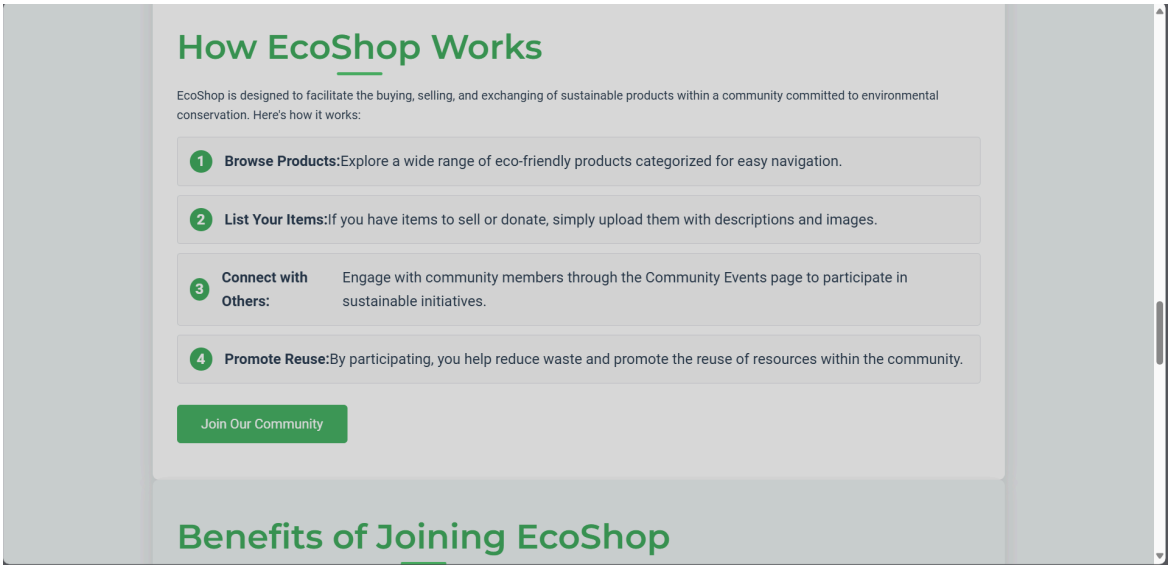
(Fig 4.2: ER diagram illustrating collections and relationships.)

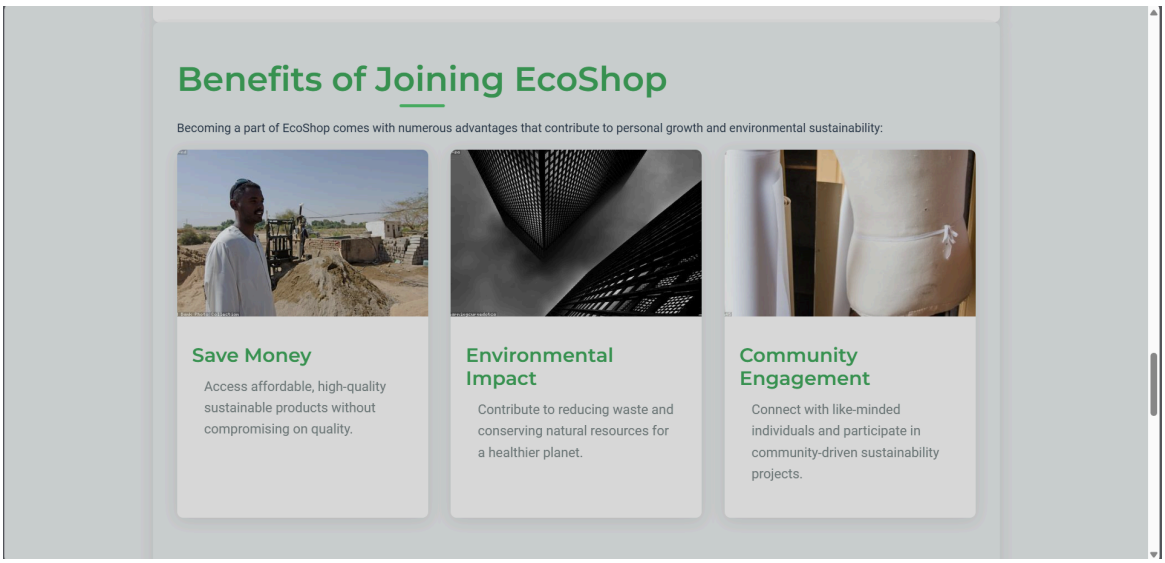
4.3 USER INTERFACE DESIGN

- **Homepage:** Introduction to the platform and its purpose.
- **Image Upload Page:** Allows users to select and upload images.
- **Results Page:** Displays classification outcomes with additional information.
- **Educational Resources:** Provides guidance on waste management practices.




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4 Promote Reuse:By participating, you help reduce waste and promote the reuse of resources within the community.
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Benefits of Joining EcoShop


Benefits of Joining EcoShop
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Community Engagement
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Product Name | Price | Place

Product Image

Choose File No file chosen

Upload Product



plastic bottle

Price: ₹10.00

Place: chennai

Category: under processing category

Owner: hari

Contact Email:
harichselvamc@gmail.com

Edit Delete



plastic bottle

Price: ₹40.00

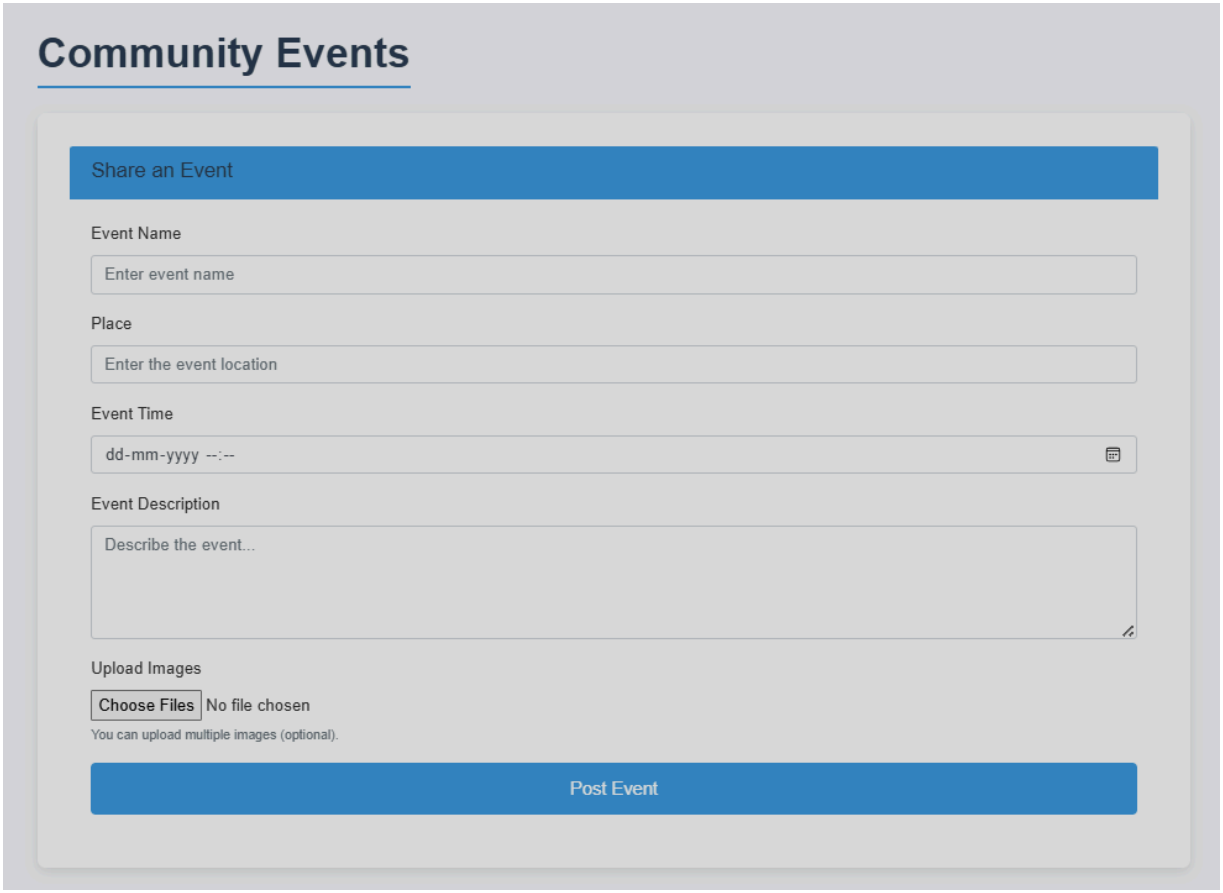
Place: chennai

Category: under processing category

Owner: hari

Contact Email:
harichselvamc@gmail.com

Edit Delete





```
    query = {
        "image": {"$exists": True, "$ne": None},
        "category": {"$in": ["under processing category"]}
    }
    products = collection.find(query)

    for product in products:
        product_id = product['_id']
        try:
            # Decode the base64 image
            img_data = base64.b64decode(product['image'])
            img = Image.open(io.BytesIO(img_data))
            img = img.convert("RGB")

            # Preprocess the image and make a prediction
            img_array = preprocess_image(img, IMG_WIDTH, IMG_HEIGHT)
            predicted_class = predict_image(img_array)

            # Update MongoDB with the predicted category
            update_product_category(product_id, predicted_class)

        except Exception as e:
            print(f"Error processing product ID {product_id}: {e}")

if __name__ == '__main__':
    process_and_update_products()
```

```
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metrics` will be empty until you train or evaluate the model.
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 81ms/step
Product ID 671d1e8187cf34cb41f3aa42 updated with category 'recycle'
```

CHAPTER 5: IMPLEMENTATION

5.1 BACKEND DEVELOPMENT

5.1.1 Flask Framework

- **Routing:** Defines endpoints for home, upload, and results pages.
- **Templates:** Utilizes Jinja2 for dynamic content rendering.
- **Error Handling:** Implements error pages for invalid inputs or system failures.

5.1.2 RESTful APIs

- **Endpoints:**
  - **/upload:** Accepts POST requests with image data.
  - **/classify:** Processes images and returns classification results.
- **Data Formats:** Uses JSON for data interchange between client and server.

5.2 MACHINE LEARNING MODEL

5.2.1 Dataset Preparation

- **Data Sources:** Images collected from open datasets like TrashNet.
- **Classes:** Two primary classes - recyclable and biodegradable.
- **Preprocessing:**
  - Image resizing and normalization.
  - Data augmentation techniques to improve model robustness.

5.2.2 Model Architecture

- **CNN Layers:**
  - **Convolutional Layers:** Extract features from images.
  - **Pooling Layers:** Reduce spatial dimensions.
  - **Fully Connected Layers:** Interpret features for classification.
- **Activation Functions:** ReLU for hidden layers, Softmax for output layer.
- **Optimization:** Adam optimizer with a suitable learning rate.

Figure 5.1: CNN Model Architecture

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 78, 78, 32)	896
max_pooling2d (MaxPooling2D)	(None, 39, 39, 32)	0
conv2d_1 (Conv2D)	(None, 37, 37, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 18, 18, 64)	0
flatten (Flatten)	(None, 20736)	0
dense (Dense)	(None, 128)	2,654,336
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 2)	258

(Fig 5.1: Diagram of the CNN layers and connections.)

5.2.3 Training and Validation

- **Training Parameters:**
  - **Epochs:** Number of full training cycles.
  - **Batch Size:** Number of samples per gradient update.
- **Validation Split:** Portion of data reserved for validation.
- **Performance Metrics:** Accuracy, precision, recall, F1-score.

Table 5.1: Model Performance Metrics

Metric	Training Set	Validation Set
Accuracy	95%	93%
Precision	94%	92%
Recall	96%	93%
F1-Score	95%	92.5%

5.3 DATABASE MANAGEMENT

Effective database management is crucial for the smooth operation of the waste classification system, ensuring data integrity, security, and performance.

5.3.1 MongoDB Integration

CONNECTION HANDLING:

Secure connections to the MongoDB database are established using authentication protocols and SSL encryption. This ensures that all data transmitted between the application and the database remains confidential and secure, protecting against unauthorized access and data breaches.

CRUD OPERATIONS:

Create, Read, Update, and Delete (CRUD) operations are implemented using the PyMongo library, which facilitates easy interaction with the MongoDB database from the Python application. This enables efficient management of user data, image uploads, and classification results.

INDEXING:

Indexes are created on key fields, such as timestamps and waste category types, to enhance the speed of queries. By indexing these fields, the system can retrieve data more quickly, improving overall performance and user experience, especially during data retrieval operations.

5.3.2 DATA STORAGE STRATEGIES

IMAGE STORAGE:

Uploaded images are stored as base64-encoded strings in the MongoDB database. This method allows for efficient storage and retrieval of image data while keeping the database lightweight. Base64 encoding also simplifies the handling of images within the application, as they can be easily embedded in JSON responses.

RESULT CACHING:

Recent classification results are cached to improve performance and reduce response times. By storing frequently accessed data in memory, the application can quickly provide results for repeat queries, enhancing user experience, particularly for users who may upload similar waste types.

BACKUP AND RECOVERY:

Regular backups of the database are performed to prevent data loss due to system failures or accidental deletions. Automated backup solutions are implemented to ensure that data can be restored quickly and effectively in case of an emergency, maintaining data integrity and reliability.

CHAPTER 6: TESTING AND EVALUATION

6.1 TESTING STRATEGIES

- **Unit Testing:**
  - Each function and module of the application was tested individually to ensure they perform as expected. This helped identify bugs at an early stage and confirmed that each unit of code meets its design specifications.
- **Integration Testing:**
  - After unit testing, modules were tested together to verify their interactions. This step ensured that data flows correctly between components and that integrated parts of the system function seamlessly as intended.
- **System Testing:**
  - The complete system was validated against the specified requirements. This comprehensive testing phase confirmed that the application meets all functional and non-functional requirements and behaves correctly in different scenarios.
- **User Acceptance Testing (UAT):**
  - Feedback was gathered from potential users to assess the application's usability and functionality. This stage involved real users interacting with the system to validate that it meets their needs and expectations before final deployment.

6.2 PERFORMANCE EVALUATION

- **Load Testing:**
  - Simulations of multiple concurrent users were conducted to assess the system's scalability. This testing evaluated how the application performs under high user loads, ensuring it can handle expected traffic without degradation of service.
- **Response Time:**
  - The average time taken for the system to classify images was measured. This metric is critical for user experience, as it reflects how quickly users receive feedback on their uploads.
- **Accuracy Testing:**
  - The machine learning model was evaluated using unseen test data to determine its classification accuracy. This testing ensures that the model generalizes well to new data and performs reliably in real-world scenarios.

Table 6.1: System Performance Metrics

Test Type	**Result**
Load Handling	Up to 1000 users/hour
Average Response Time	1.2 seconds
Classification Accuracy	93%

6.3 USER FEEDBACK

- **Ease of Use:**
  - Users reported that the interface is intuitive and easy to navigate, enhancing their overall experience with the application.
- **Educational Value:**
  - Users appreciated the additional information provided on waste disposal methods, which increased their understanding and awareness of environmental sustainability.
- **Improvement Areas:**
  - Users suggested adding more waste categories to broaden the classification options, indicating a desire for a more comprehensive system that addresses a wider range of waste types.

CHAPTER 7: CONCLUSION AND FUTURE WORK

The project successfully developed a machine learning-based system to classify waste products as recyclable or biodegradable using image analysis. It addresses critical challenges in waste management by automating the sorting process and engaging the public.

Future Enhancements:

- **Expand Classifications:** Include more waste categories like hazardous waste.
- **Mobile Application:** Develop mobile apps for wider accessibility.
- **Localization:** Support multiple languages and regional waste management guidelines.
- **Integration with Municipal Services:** Collaborate with local authorities for waste collection services.

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